(Please write your Enrolment No. immediately) Enrolment No._____

MID TERM EXAMINATION
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
VIII[th] Semester, April, 2024

Paper Code: ETEC-406 **Subject: Ad Hoc and Sensor Networks**

Time :- $1\frac{1}{2}$ Hrs. **Max. Marks: 30**

Note: Attempt Q. No.1 which is compulsory and any two more questions from remaining.

Q.No.	Questions	Max. Marks
1 (a)	How does packet queuing mechanism of MACA differ from MACAW?	2
1 (b)	How replay attack can be prevented?	2
1 (c)	Explain role of routing protocol in provisioning of QoS guarantees for ad hoc wireless networks.	2
1 (d)	Why table-driven routing protocol is not suitable for high mobility environments?	2
1 (e)	Explain impact of failure of proxy nodes in split-TCP.	2
2 (a)	List major advantages of ad hoc wireless internet.	4
2 (b)	Which protocol is more bandwidth efficient, RTMAC or MACA/PR? Explain in detail.	6
3 (a)	Discuss differences in topology reorganization in DSDV and CGSR routing protocols.	5
3 (b)	Discuss pros and cons of assigning responsibility of end-to-end reliability to application layer.	5
4 (a)	Explain key encrypting key method.	4
4 (b)	How inherent property of wireless ad hoc network introduce difficulties during implementation of security in routing protocols? Explain in detail.	6

(Please write your Exam Roll No.) *Exam Roll No. .....*

ANNUAL EXAMINATION

EIGHTH SEMESTER [B.TECH.] JULY-2023

Paper Code: ETEC-406 ***Subject: Ad Hoc and Sensor Networks***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No1 which is compulsory. Select one question from each unit.

Q1 Attempt All:

(a) What is an ad-hoc network? Why ad hoc networks are needed? Discuss. **(4)**

(b) What is Sensor Network Localization. **(4)**

(c) Explain goals to be achieved in transport layer protocol for ad hoc wireless networks. **(5)**

(d) List the characteristics of ideal routing protocol for ad hoc wireless network. **(4)**

(e) Define data relaying in a WSN. **(4)**

(f) What is data dissemination in a wireless sensor network? **(4)**

UNIT-I

Q2 (a) Explain about the Contention-based MAC protocols with scheduling mechanism. **(6.5)**

(b) Outline the design challenges in mobile adhoc networks and wireless sensor network. **(6)**

Q3 (a) What are the advantages of reservation based MAC protocols over contention based MAC protocols? **(6.5)**

(b) Explain the issues in designing the MAC protocols in ad hoc networks and give the classification of MAC protocols. **(6)**

UNIT-II

Q4 (a) What is the need for power management in ad hoc network? Discuss the approaches for power aware routing protocol. **(6)**

(b) Give classification of routing protocols in ad hoc networks. Explain any two. **(6.5)**

Q5 (a) Why TCP protocols used in wired network is not suitable for wireless networks? **(6.5)**

(b) What do you understand by network security requirements? Discuss the issues and challenges in secure routing of ad hoc wireless networks. **(6)**

P.T.O.